

Technical Notes

Feed-Forward Compensation of Body-Pointing Uncertainties for Laser Communication Terminals

René Rüddenklau,^{*} Salvatore Barone,[†]

Elisa Garbagnati,[‡] and Simon Spier[§]

German Aerospace Center (DLR), 82234 Weßling, Germany
and

Georg Schitter[§]

Technical University Wien, 1040 Vienna, Austria

<https://doi.org/10.2514/1.G009261>

Nomenclature

A	=	gain error matrix, a.u.
b	=	calibrated offset rate vector, deg·s ⁻¹
C	=	calibrated gain error matrix, a.u.
d	=	phase, rad
f	=	frequency, rad · s ⁻¹
G	=	amplitude, rad · s ⁻¹
$I_{3 \times 3}$	=	identity matrix, a.u.
M	=	magnification, a.u.
o	=	offset rate vector, deg·s ⁻¹
q	=	quaternion, a.u.
$\hat{q}$	=	estimated quaternion, a.u.
q_{err}	=	pointing error quaternion, a.u.
$[q_{1:3}]_{\times}$	=	quaternion cross-product, a.u.
R	=	rotation matrix, a.u.
r	=	inertial reference direction, a.u.
t	=	time, s
u	=	direction in body frame, a.u.
$\hat{u}$	=	estimated direction in body frame, a.u.
α	=	pointing error angle, deg
e	=	attitude error vector, deg
θ	=	optical terminal (Euler) angles, deg
Λ	=	attitude direct cosine matrix, a.u.
$\hat{\Lambda}$	=	estimated attitude direct cosine matrix, a.u.
$\tilde{\Lambda}$	=	attitude error direct cosine matrix, a.u.
μ_{θ}	=	mean of optical terminal (Euler) angles, deg
ν	=	sensor noise vector, deg·s ⁻¹
$\Xi(q)$	=	quaternion matrix, a.u.
σ_{θ}	=	standard deviation of optical terminal (Euler) angles, deg
σ_{FOU}	=	field of uncertainty standard deviation, deg

ϕ	=	true attitude determination and control system pointing angles, deg
$\hat{\phi}$	=	gyroscope propagated angles, deg
ψ	=	combined laser communication terminal and attitude determination and control system (Euler) angles, deg
ω	=	true rate vector, deg·s ⁻¹
$\tilde{\omega}$	=	measured rate vector, deg·s ⁻¹
$\hat{\omega}$	=	calibrated rate vector, deg·s ⁻¹

I. Introduction

FREE-SPACE optical communication is achieved by low-divergent laser beams, which require precise pointing. Precise host platform pointing is required, especially for a CubeSat-sized laser communication terminal (LCT), which often waives the use of a dedicated hemispherical pointing assembly. Moreover, a high degree of pointing stability is necessary to facilitate the successful acquisition process for intersatellite links. In general, the suppliers of the satellite bus and the laser communication terminal are two distinct entities [1]. Consequently, valuable information regarding the attitude is often not shared at a sufficient update rate between the involved systems.

Known efforts to make use of exchanging data about attitude knowledge are presented in recent works [2,3]. A common scenario is based on the relative attitude information of the optical payload with respect to a beacon as an additional input for the spacecraft's Kalman attitude estimation filter. This enables sub-arcsecond pointing accuracy and increased stability during an optical link [4]. It is especially useful when the satellite experiences highly dynamic scenarios such as zenith overflights during a direct-to-Earth (DTE) link, pointing toward an optical ground station (OGS), or in an intersatellite link (ISL) when the relative orientation to the other satellite has to be maintained with very high accuracy. In contrast, the approach proposed in this work targets primarily scenarios where the link is not yet established, the so-called acquisition phase.

For this phase, a common approach to compensate for instabilities is the use of secondary sensors such as gyroscopes within the attitude control system [1]. A general usage of gyroscope sensors for stabilization is known from camera-stabilizing gimbals or free-space optical gimbals. One method uses the measured drift rates to compensate for disturbances within the velocity path of a closed-loop controller in combination with an observer model [5]. However, inter-subsystem knowledge of the attitude is not used. Primarily, information within the gimbal subsystem is exploited to reach the pointing goal, relying only on one type of sensor. Another aspect of this approach is that highly accurate gyroscope sensors have to be used in order to have standalone stabilizing support for the main attitude controller.

Although such precise gyroscopes are available (e.g., fiber optic gyroscopes, which promise high sensitivities), they do not fit the size, weight, and power constraints of a CubeSat-sized LCT [6]. However, there are ongoing efforts in development approaches that target the use case of microsatellites to implement so-called integrated optical gyroscopes with commercial off-the-shelf components in a small form factor [7] or even as photonic integrated circuits [8]. In this work, however, only commercially available solutions are considered, which need to fit on a printed circuit board (PCB) in the CubeSat PC/104 standard together with the rest of the optical payload. The PCB itself will be mounted to the optomechanics, ensuring a fixed orientation between the optics and sensors. Further, it is proposed to make use of the attitude determination and control system (ADCS), which provides highly accurate information during the acquisition phase, where no optical feedback is

Received 11 April 2025; accepted for publication 13 January 2026; published online Open Access 31 March 2026. Copyright © 2026 by the American Institute of Aeronautics and Astronautics, Inc. All rights reserved. All requests for copying and permission to reprint should be submitted to CCC at www.copyright.com; employ the eISSN 1533-3884 to initiate your request. See also AIAA Rights and Permissions <https://aiaa.org/publications/publish-with-aiaa-rights-and-permissions/>.

^{*}Research Associate, Optical Satellite Links, Institute of Communications and Navigation; rene.rueddenklau@dlr.de; also Ph.D. Candidate, Mechatronics and Power Electronics Institute, Technical University Wien.

[†]Student, Optical Satellite Links, Institute of Communications and Navigation.

[‡]Research Associate, Optical Satellite Links, Institute of Communications and Navigation.

[§]Professor, Mechatronics and Power Electronics Institute.

available. The disadvantage of its relatively slow sampling rate of usually 1 Hz is then enhanced by propagation with the onboard gyroscope sensor of the LCT.

This work's main contribution proposes a concept that combines the attitude data of the ADCS and LCT secondary rate sensors and demonstrates its feasibility for augmenting attitude knowledge during pointing without optical feedback, thus mitigating body-pointing uncertainties and relative mounting inaccuracies and consequently reducing the time required for the acquisition phase. Therefore, first, state-of-the-art CubeSat platforms and the expected disturbances are analyzed in Sec. II. With this information at hand, in Sec. III, the methods for deriving the calibration between involved subsystems, alongside the formulation of relevant estimation algorithms and the details of the feed-forward compensation concept, are introduced. Finally, Sec. IV discusses the selection of the propagation algorithm by simulation and analyzes the results of the performance on a hardware-in-the-loop breadboard.

II. Intersatellite Link Scenario on 6U CubeSats

Optical links, especially between two LCTs, impose high accuracy requirements on the ADCS due to the narrow divergence of the laser beam. The following section evaluates state-of-the-art CubeSat platforms in order to quantify the challenges that must be overcome to reliably establish and maintain a laser communications link. This is relevant because CubeSats represent an emerging use case for LCTs, although they have limitations concerning their ADCS with respect to larger satellites. A 6-unit CubeSat platform is used as a reference to numerically analyze the performance improvements obtained. This includes the disturbances present in the proposed scenario. Using the given infrastructure on a CubeSat, a propagation algorithm and an attitude data exchange procedure are proposed to mitigate the identified inaccuracies.

A. Disturbance Identification

The relevant disturbances are microvibrations and the stability of the satellite's pointing resulting from the limited control accuracy between two consecutive ADCS updates [9]. Further, an initial pointing offset between the ADCS frame and the optical-axis frame, due to mounting tolerances and launch loads, may occur. Section IV.B.1 gives a solution to this calibration problem. In order to compare the improvements of the proposed algorithms for dynamic disturbances, it is necessary to understand which pointing accuracy and attitude knowledge are available on state-of-the-art CubeSat platforms [10–15]. Therefore, a small selection was made and is displayed for comparison (see Table 1). In addition, the CubeISL [16] mission is used as a reference wherever LCT design parameters are necessary for the evaluation. The same hardware prototype is used for the following hardware-in-the-loop validation measurements.

It can be seen that the measurement or estimation of the current attitude (attitude knowledge) is generally better than the accuracy achieved by the control system (attitude accuracy), as expected. It is important to note that these parameters are also dependent on the intended use case. Star trackers, which are the most accurate means of measurement available during acquisition, are more susceptible to rotations around the sensor's surface normal [1]. Additionally, light pollution can influence the quality of measurements.

Based on the aforementioned evaluation—and assuming that the pointing does not get worse than on a demonstrated 3U CubeSat—it can be assumed that the drift rate, which is equivalent to the pointing stability as referred to in Table 1, within the pointing accuracy yields $0.1 \text{ deg}\cdot\text{s}^{-1}$ in the worst case (3U CubeL). This can be assumed since larger 6U CubeSats will have improved pointing capabilities. The worst case is therefore used in the pointing, acquisition, and tracking simulations [9]. All values are understood as 3σ , and therefore a satellite requirement of $\pm 0.2 \text{ deg}$ field of uncertainty (FOU) and a knowledge of $\pm 0.1 \text{ deg}$ were chosen as a baseline, since most 6U platforms are capable of reaching these requirements [1]. In comparison, the divergence angle of the LCT is on the order of tens of millidegrees, and the field of view (FOV) is usually about 10 times as large [16].

Table 1 Brief survey of selected 6U ADCS commercial off-the-shelf satellite buses

Company / Unit	Accuracy, deg	Knowledge, deg	Stability, $\text{m deg}\cdot\text{s}^{-1}$
AAC Clyde Space / iADCS400 [10]	<1	0.008	n/a
Alén Space / 6U [11]	0.2	0.1	n/a
Blue Canyon / XB6 [12]	$\pm 0.002 (1\sigma)$	$0.003 (3\sigma) [1]$	0.27
Gomspace / 6UPremium [13]	0.07	0.056	n/a
Nano avionics / M6PMax [14]	$\pm 0.08 (1\sigma)$	$\pm 0.04 (1\sigma)$	n/a
DLR / 3U CubeL [15]	<1 (3σ)	n/a	100 (3σ)

In Fig. 1, the perturbations acting on the attitude of the satellite with respect to the optical axis of the LCT are shown at the top, on the left for a T-DAT (terminal for detection, adjustment, and tracking) and on the right for its link partner T-PAT (terminal for pointing, acquisition, and tracking) [9]. The middle row shows the unperturbed search pattern, a 3.2 s spiral for T-PAT and a 30.2 s grid search pattern for T-DAT for an FOU of $\pm 0.1 \text{ deg}$. The resulting superposition in the bottom row shows that the perturbations result in a distorted pattern from the start of the acquisition, marked with a yellow circle, to the point of link establishment, marked with a red circle. All results are based on an acquisition simulation for an optical ISL between CubeSats [9]. When looking at an increased FOU, such as the envisaged $\pm 0.2 \text{ deg}$, which is beyond the FOV of the terminal, the likelihood of successful acquisition drops drastically [9]. The proposed concept aims to propagate with enough precision to identify pointing stability errors that are beyond the

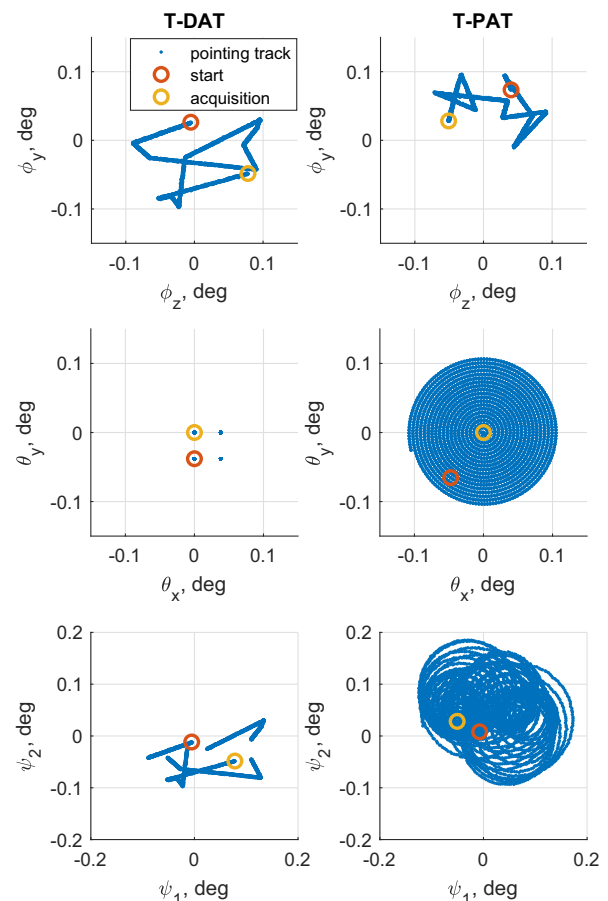


Fig. 1 Example of uncompensated search patterns being deformed due to limited body-pointing precision.

accuracy of body pointing but within the capabilities of the optical terminal's fine-steering mirror (FSM).

B. Attitude Propagation and Compensation Concept

A standard CubeSat platform consists of the mechanical structure that defines the available units and the functionality provided by various subsystems. The ADCS and corresponding 6U buses can be purchased from various satellite manufacturers, allowing the user to focus on the payloads. To date, in most cases, the subsystems have been developed and operated independently. The proposed concept provides for an exchange of information between the individual subsystems in such a way that information can be shared and is commonly available even when multiple LCTs are used. A conventional ADCS operates with high-level attitude control of about 0.1–10 Hz [17]. Thus, information from the estimation algorithm of the ADCS, which is usually a Kalman filter, can be used to reduce the FOU from the full field of regard (FOR) of the optical system to the actual FOU.

It needs to be considered that the search pattern as shown above in Fig. 1 of the LCT is sampled faster—with sampling rates above those of the ADCS—on the order of several hundred to a thousand times per second, since the period of a scan trajectory is directly related to the expected time needed for acquisition. Therefore, it is necessary that the update rate of the estimator matches the update rate of the optical control system. Following this train of thought, a separate estimation algorithm using gyroscope-measurement-based propagation is proposed for the control unit of the LCT, which is initialized with the estimation value of the ADCS. It is assumed that this system will generally provide more accurate information since the ADCS utilizes the aforementioned star tracker.

However, instead of only extrapolating this initial value, the LCT makes use of its own sensor consisting of a microelectromechanical system (MEMS) gyroscope sensor. This is done until the next update of the ADCS data is available. The assumption that this approach improves the attitude knowledge of the LCT is therefore investigated in this work. When comparing the two data sources, it should be noted that the sensors involved must be calibrated beforehand to align their reference frames. It should be noted that this approach is not exclusive to the use of gyroscope sensors. Alternative sensors with sufficient sampling rates, such as magnetometers, may also be employed. The methods, including the estimation and calibration algorithms, are described in more detail in Sec. III.

Figure 2 shows the commanding of the FSM and where the propagated information is inserted. The FSM is steered in both the acquisition phase and the tracking phase. In the latter case, the control loop is closed via the feedback from the quadrant photodiode (QPD) to the tracking controller. However, during acquisition, a previously defined search pattern is executed until a beacon is detected following an acquisition scheme [9]. In order to include the measured deviation from the ideal attitude, the pointing error is added to the output of the search pattern so that the search pattern remains distortion-free with respect to the QPD's inertial frame.

III. Sensor Calibration and Attitude Augmentation

Sensors are subject to disturbances and errors, each influencing the measurements. Therefore, it is necessary to evaluate the calibration and measurement accuracy of the involved systems. Additionally, it is described how the propagation algorithm is implemented,

which aims to augment the a priori attitude knowledge of an LCT before the establishment of a laser link.

A. Calibration Algorithm

Each calibration error can be described as either a gain error or an additive offset. For a three-dimensional vector of true values ω , the gain components make up matrices $A \in \mathbb{R}^{3 \times 3}$ influencing the scaling of each axis A_s , modeling cross-axis effects represented by nonorthogonality A_{no} as well as mounting errors A_m [18].

Equation (1) displays how the measured vector $\tilde{\omega}$ is obtained by adding the offset vector o and the sensor noise ν :

$$\tilde{\omega} = \underbrace{(A_{no} A_s A_m)}_A \cdot \omega + o + \nu \quad (1)$$

A calibration algorithm aims to determine the influence of the matrix A and the offset o , while the sensor noise cannot be accounted for. Therefore, the calibrated vector $\hat{\omega}$ is influenced by the measurement noise from $\tilde{\omega}$, while $C = A^{-1}$ and $b = -o$. Thus, Eq. (2) shows how a measurement is transformed after the 12 calibration parameters have been determined:

$$\hat{\omega} = C \cdot (\tilde{\omega} + b) \quad (2)$$

1. Static Calibration of the Gyroscope Sensor

The calibration of the gyroscope sensor uses the reference value of the satellite's ADCS to adjust the measurements. This is done using a standard least-squares method (LSM) [19]. It is important that the measurements include motion at various velocities and on each axis to obtain acceptable results. Otherwise, overfitting to limited sample sets may occur, which can ultimately lead to erroneous results. MEMS gyroscope sensors suffer from the mentioned bias errors as well as gain errors and, additionally, nonorthogonality but usually do not get damaged by launch vibration loads; therefore, a preflight calibration can be used as a reference for an on-orbit calibration [20]. The result is required for the following processing steps and is shown in Sec. IV.B.1.

2. Online Bias Calibration of the Gyroscope Sensor

The LSM is a batch-type static procedure, which means that a data set must be obtained by rotating the satellite beforehand. Since the quality of the calibration is highly dependent on the variety of platform motions during those measurements as well as the gyroscope's susceptibility to temperature changes that induce bias drifts, the calibration has to be verified during early operations. In order to compensate for ever-changing environmental conditions, a recursive bias estimation algorithm is proposed to dynamically adapt to changes affecting sensor readings.

Taking into account the proposed system architecture (see Fig. 2), the satellite ADCS will provide the terminal with its precise estimate of the attitude quaternion. Using this information, two successive attitudes can be used to estimate the true angular velocity ω of the laser terminal. This is either directly derived from the ADCS or by the derivative indicated by the Laplace variable s in Fig. 3. The bias can be estimated with Eq. (3) by comparing this estimated angular velocity with an average of the measurements from the onboard gyroscope sensor between two ADCS updates. As a result, the

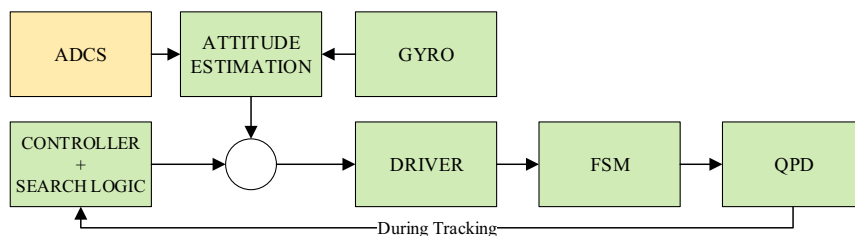


Fig. 2 Feed-forward compensation scheme using the data of the ADCS and the propagated information from the onboard gyroscope sensor.

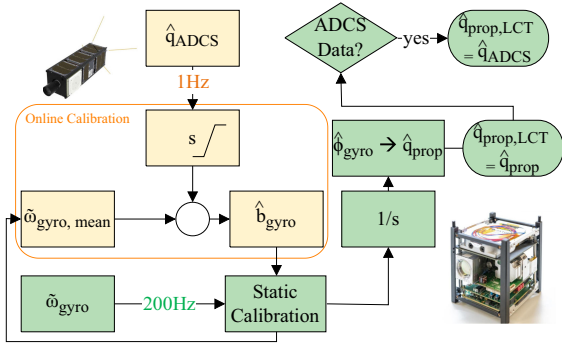


Fig. 3 Quaternion propagation algorithm scheme to fuse data from the ADCS and the LCT's gyroscope sensor, sampled at 1 Hz (orange) and 200 Hz (green), respectively.

current measurement can be calibrated with the updated parameter in Eq. (2):

$$b_{\text{gyro}} = \omega_{\text{ADCS}} - \hat{\omega}_{\text{gyro, mean}} \quad (3)$$

However, this online calibration process has to be used with care. Scenarios with higher accelerations may lead to wrong averages of the transmitted satellite mean angular velocity. Therefore, the averaging should take several samples into account but only when the velocity change between two consecutive updates does not exceed a predefined threshold. This threshold can be defined by simulating changing acceleration conditions and is presented in Sec. IV.A.

B. Formulation of the Proposed Attitude Estimation Algorithms

After the calibration routines are in place, it is possible to develop the estimation algorithm for attitude propagation. The aim of a computationally efficient algorithm is to improve the attitude accuracy of the LCT between two successive updates from the satellite's ADCS [21]. Two estimation strategies have then been considered and will be presented in the following. The algorithms will use the attitude information, in the form of quaternions, from the attitude estimation system and either the measurements from the LCT's sensor or internal propagation to provide accurate attitude knowledge at high sampling rates.

For the following example in Sec. IV, it is assumed that the satellite ADCS operates at 1 Hz while the LCT terminal has a sampling frequency of 200 Hz to account for higher-frequency disturbances like microvibration and body-pointing drift. According to these rates, the estimated quaternion vector, representing the attitude of the LCT, is updated every 1 s with the estimated value of the quaternion vector of the ADCS knowledge $\hat{q}_{\text{ADCS}}$. Between updates, the algorithms estimate the attitude of the LCT, using either linear extrapolation or quaternion kinematics propagation. The first proposed strategy relies only on the attitude information given by the ADCS updates. The latter algorithm also applies the additional LCT gyroscope sensor measurements.

As a first approach, a linear extrapolation of the quaternions between the ADCS updates at the LCT's sampling rate of 200 Hz is pursued. As shown in Eq. (4), this algorithm performs the extrapolation of the predicted quaternion vector $\hat{q}_{\text{extrap}}$ by deriving it according to a linear model from the two previously computed quaternions, with a step size of 5 ms. Meanwhile, every second the value of $\hat{q}_{\text{extrap}}$ is reset to the attitude information $\hat{q}_{\text{ADCS}}$, broadcast by the satellite.

Since linear extrapolation does not preserve the unit norm of the quaternion, the estimate is renormalized as in Eq. (5):

$$\hat{q}_{\text{extrap}}(t_k + n\Delta t) = \hat{q}_{\text{ADCS}}(t_k) + \frac{n\Delta t}{t_k - t_{k-1}}(\hat{q}_{\text{ADCS}}(t_k) - \hat{q}_{\text{ADCS}}(t_{k-1}))$$

where $\Delta t = 5 \text{ ms}$, $n = 1, 2, \dots, 200$ (4)

$$\hat{q}_{\text{extrap}}(t_k + n\Delta t) \leftarrow \frac{\hat{q}_{\text{extrap}}(t_k + n\Delta t)}{\|\hat{q}_{\text{extrap}}(t_k + n\Delta t)\|} \quad (5)$$

The other proposed algorithm relies on the propagation of the quaternions by discrete-time integration of their kinematics [22] as shown in Eq. (6):

$$\dot{q}(t) = \frac{1}{2} \Xi(q(t)) \omega(t) \quad (6)$$

where $\omega \in \mathbb{R}^{3 \times 1}$ is the angular velocity vector and the quaternion matrix $\Xi(q(t))$ is computed from the quaternion cross-product $[q_{1:3}]_{\times}$ as follows [22]:

$$\Xi(q(t)) \equiv \begin{bmatrix} q_4 I_{3 \times 3} + [q_{1:3}]_{\times} \\ -q_{1:3}^T \end{bmatrix} = \begin{bmatrix} q_4 & -q_3 & q_2 \\ q_3 & q_4 & -q_1 \\ -q_2 & q_1 & q_4 \\ -q_1 & -q_2 & -q_3 \end{bmatrix} \quad (7)$$

Contrary to the linear extrapolation method, the quaternions are estimated by exploiting their own kinematics law instead of making use of a pure mathematical linear relation. With this approach, it is possible to take into account the natural time evolution of the considered quaternions and to combine it with the information coming from the angular velocity estimates of the ADCS. The proposed propagation, referred to as quaternion propagation, also exploits the LCT measurements for the discrete-time integration. Indeed, it uses the satellite updates $\hat{q}_{\text{ADCS}}$ and the angular velocity $\hat{\omega}_{\text{gyro}}$ measured from the MEMS gyroscope sensor in $\dot{q}(t) = (1/2)\Xi(q(t))\omega(t)$. The algorithm scheme is illustrated in Fig. 3. The propagation $\hat{q}_{\text{prop,LCT}}$ occurs at a higher frequency of 200 Hz, as a result of the integration $1/s$ of the gyroscope measurements $\hat{\phi}_{\text{gyro}}$, and the error drift is kept bounded thanks to the ADCS quaternions periodically received at 1 Hz. The flow chart also includes the online calibration method described by Eqs. (2) and (3), which discards measurements indicated by the saturation symbol after the Laplace variable s beyond a set velocity threshold to ensure the quality of the online calibration.

C. Feed-Forward Fine-Steering Mirror Control

Figure 4 illustrates how the estimated quaternions are transformed to control the FSM. It starts with the attitude knowledge of the LCT and ends with the required driving command voltage to achieve the necessary optical compensation angle.

First, the pointing error is determined by multiplying the propagated quaternion with the target quaternion to retrieve the pointing error quaternion $q_{\text{err}} = \hat{q} \otimes \bar{q}$. The details of this operation are reported in the Appendix. The attitude matrix Λ of the LCT in the hexapod reference frame is derived from the quaternions [22] according to Eq. (8):

$$\Lambda(q_{\text{err}}) = \begin{bmatrix} q_1^2 - q_2^2 - q_3^2 + q_4^2 & 2(q_1 q_2 + q_3 q_4) & 2(q_1 q_3 - q_2 q_4) \\ 2(q_2 q_1 - q_3 q_4) & -q_1^2 + q_2^2 - q_3^2 + q_4^2 & 2(q_2 q_3 + q_1 q_4) \\ 2(q_3 q_1 + q_2 q_4) & 2(q_3 q_2 - q_1 q_4) & -q_1^2 - q_2^2 + q_3^2 + q_4^2 \end{bmatrix} \quad (8)$$

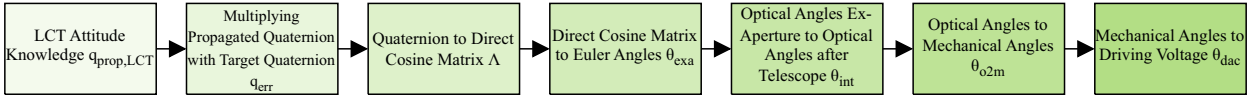


Fig. 4 Transformation from derived attitude quaternions to FSM control command voltage.

Afterward, from the obtained direct cosine matrix, the sequence $\{\theta_x^3, \theta_y^2, \theta_z^1\}$ of hexapod Euler angles is computed [22] as

$$\begin{cases} \theta_{x,hxp} = \text{atan2}(2(q_2q_3 + q_1q_4), 1 - 2(q_1^2 + q_2^2)) &= \text{atan2}([\Lambda]_{23}, [\Lambda]_{33}) \\ \theta_{y,hxp} = \arcsin(-2(q_1q_3 - q_2q_4)) &= \sin^{-1}(-[\Lambda]_{13}) \\ \theta_{z,hxp} = \text{atan2}(2(q_1q_2 + q_3q_4), 1 - 2(q_2^2 + q_3^2)) &= \text{atan2}([\Lambda]_{12}, [\Lambda]_{11}) \end{cases} \quad (9)$$

Comparing the reference frames of the FSM and of the hexapod illustrated in Fig. 5, it is noticeable that a rotation around $\theta_{x,fsm}$ corresponds to a rotation around $\theta_{z,hxp}$ and analogously $\theta_{y,fsm}$ to $\theta_{y,hxp}$ as well as $\theta_{z,fsm}$ to $-\theta_{x,hxp}$. This is equally expressed by

$$\theta_{x,fsm} = \theta_{z,hxp}, \quad \theta_{y,fsm} = \theta_{y,hxp}, \quad \theta_{z,fsm} = -\theta_{x,hxp} \quad (10)$$

where the FSM angle around the third axis has been canceled because rotations around the laser pointing direction axis do not affect the pointing. The known pointing error is therefore given in the FSM reference frame. However, to command a corresponding voltage to the FSM control voltage in a feed-forward manner, it is necessary to convert from the optical domain ex-aperture via the internal optical domain after the compression telescope to the mechanical domain and finally to the electrical domain as follows: $\theta_{x/y,exa} \rightarrow \theta_{x/y,int} \rightarrow \theta_{x/y,o2m} \rightarrow \theta_{x/y,dac}$, where $\theta_{x/y,exa} = \theta_{x/y,fsm}$. In a first step, the conversion accounting for the telescope magnification $M_{telescope}$ is performed using Eq. (11):

$$\theta_{x/y,int} = \theta_{x/y,exa} \cdot M_{telescope} \quad (11)$$

In the following, the conversion from the optical to the mechanical domain needs to be performed. In general, an optical beam is reflected with the same angle as the angle of incidence, therefore doubling the mechanical deflection that is used for compensation. However, because the FSM is often mounted at a 45 deg angle with respect to the incoming beam in many optical designs, three-dimensional angle contemplation is necessary. Due to the coupled axes, a rotation of the mirror

$$\begin{aligned} \theta_{x,int} = & \arcsin(2 \sin(2\theta_{x,stat}) \cos(\theta_{x,o2m})^2 \cos(\theta_{y,o2m})^2 \\ & + 2 \cos(2\theta_{x,stat}) \sin(\theta_{x,o2m}) \cos(\theta_{x,o2m}) \cos(\theta_{y,o2m})^2 \\ & - \sin(2\theta_{x,stat})) \end{aligned} \quad (12)$$

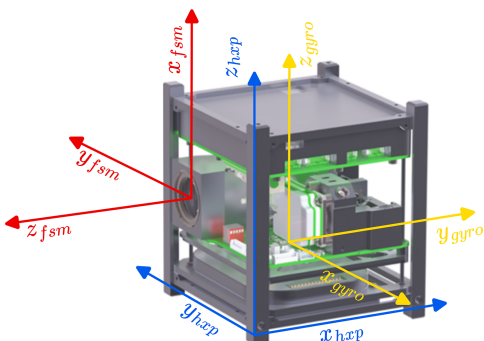


Fig. 5 Reference frames of the FSM (—), satellite/hexapod (—), and gyroscope sensor (—) in the example of the CubeISL payload [16].

has to be treated with respect to the FSM inertial system. Due to the 45 deg static mounting angle θ_{stat} , however, Eq. (12) simplifies to

$$\theta_{x,int} = \arcsin(2 \cos(\theta_{x,o2m})^2 \cos(\theta_{y,o2m})^2 - 1) \quad (13)$$

while the other axis is independent of the static angle:

$$\theta_{y,int} = \arcsin(2 \cos(\theta_{x,o2m}) \cos(\theta_{y,o2m}) \sin(\theta_{y,o2m})) \quad (14)$$

Equation (14) can then be solved for

$$\theta_{x,o2m} = \arccos\left(\frac{\sin(\theta_{y,int})}{2 \cos(\theta_{y,o2m}) \sin(\theta_{y,o2m})}\right) \quad (15)$$

which, however, still depends on the results of $\theta_{y,o2m}$. Therefore, when Eq. (15) is used as a substitute for $\theta_{x,o2m}$ in Eq. (13), $\theta_{y,o2m}$ can be retrieved after rearrangement:

$$\theta_{y,o2m} = \arcsin\left(\frac{\sin(\theta_{y,int})}{\sqrt{2 + 2 \sin(\theta_{x,int})}}\right) \quad (16)$$

As a final step, Eq. (16) can be inserted into Eq. (15) and yields after simplification

$$\theta_{x,o2m} = \arccos\left(\frac{\sqrt{2 + 2 \sin(\theta_{x,int})}}{2 \sqrt{1 - \frac{\sin(\theta_{y,int})^2}{2 + 2 \sin(\theta_{x,int})}}}\right) \quad (17)$$

With both mechanical angles at hand, the command voltage for the specific device needs to be retrieved. Since the FSM is driven in an open-loop manner during acquisition and the MEMS FSM is characterized by a nonlinear relationship between angle and voltage [23], the command is approximated by a third-order polynomial of the form

$$\theta_{x/y,dac} = a_0 + a_1 \cdot \theta_{x/y,o2m} + a_2 \cdot \theta_{x/y,o2m}^2 + a_3 \cdot \theta_{x/y,o2m}^3 \quad (18)$$

This transformation and its coefficients need to be adapted to the specific MEMS FSM and electronics in use.

IV. Validation of the Pointing Stability Compensation Algorithm

In this section, the results of the proposed propagation algorithms are presented and discussed. Based on a simulation model, all relevant methods can be compared under controlled conditions. This evaluation is the basis for the implementation of the propagation used for hardware validation. It demonstrates the advantages of the proposed concept and explores the limits for its application. Therefore, after a calibration and validation campaign, the final performance is evaluated on a hardware-in-the-loop testbed.

A. Simulation Analysis of the Attitude Knowledge Performance

The algorithms have been implemented in a Simulink environment to analyze their performance and to quantify the improvement obtained by using the additional attitude information from the LCT at 200 Hz compared to considering only the attitude updates received periodically from the satellite at 1 Hz.

For simulation purposes, reference attitude parameters have been created to represent the attitude of the satellite platform. The components of the satellite angular velocity ω_{ADCS} are implemented as damped sinusoidal functions

$$\omega(t) = G \cdot e^{-d \cdot t} \cdot \sin(f \cdot t + \Phi) \quad (19)$$

with initial amplitude gain G of $0.2 \text{ rad} \cdot \text{s}^{-1}$, f of $0.1 \text{ rad} \cdot \text{s}^{-1}$, and Φ of $[0, 5, -5] \text{ rad}$, respectively, for the three body axes and a damping coefficient d of 0.01 . The reference angular velocity is damped in order to simulate two different mission scenarios. The first scenario is a dynamic maneuver that occurs in the first half of the simulation. This is followed by a more stable attitude, which is similar to the tracking phase in the second half of the simulation.

The reference quaternion vector q_{ADCS} is then retrieved from the angular velocity according to the quaternion kinematics equation with initial conditions $q_{\text{ADCS},0} = [0; 0; 0; 1]$ and a simulation sampling rate of $1 \cdot 10^{-4} \text{ s}$. These attitude data represent the real reference quaternion of the simulation and are the input to the LCT to perform its own attitude propagation. The model of the gyroscope sensor is derived according to the data sheet (IAM20380, TDK InvenSense [24]), considering a natural frequency of 1000 Hz and a cutoff frequency of 177 Hz . The characteristic noise power of $0.02^2 \text{ dps}^2/\text{Hz}$ was measured on the test hardware.

The simulation runs for 800 s , which allows enough time to evaluate the dynamic maneuver and the tracking scenario. To quantify the estimation errors of the presented methods, an analysis of the attitude error vector ϵ in the body frame is performed to compute the pointing errors resulting from the estimation algorithms themselves. To achieve this, the estimated $\hat{\Lambda}$ and the actual Λ_{ADCS} posture matrices are first derived from $\hat{q}$ and q_{ADCS} , respectively, according to Eq. (8) but using the estimated quaternion instead of the error quaternion. Next, the attitude error matrix $\tilde{\Lambda}$ is computed as in Eq. (20), and the attitude error vector is actually extracted from its components according to the small-angle approximation [22] as $\epsilon = [\tilde{\Lambda}_{23}; \tilde{\Lambda}_{31}; \tilde{\Lambda}_{12}]$:

$$\tilde{\Lambda} = \hat{\Lambda} \Lambda_{\text{ADCS}}^T \quad (20)$$

Moreover, the pointing error α is computed for both methods, as the angle between the true direction and the estimated direction. By selecting the spacecraft boresight direction $r = [0; 0; 1]$ as the desired target direction in the inertial frame, the reference true direction u_{true} and the estimated direction $\hat{u}$ in the body frame are, respectively, obtained as

$$u_{\text{true}} = \Lambda_{\text{ADCS}} r, \quad \hat{u} = \hat{\Lambda} r \quad (21)$$

Indeed, the pointing error α is retrieved according to Eq. (22) and then converted from radians to degrees [22].

$$\alpha = \cos^{-1}(u_{\text{true}}^T \hat{u}) \quad (22)$$

To compare the algorithms, the root-mean-squared errors (RMSEs) of ϵ and α are computed for the entire simulation time

span (here referred to as “Full simulation”), for the first 400 s (the “Dynamic scenario”), and over the last part of the simulation (referred to as the “Tracking scenario”). Subsequently, the vectorial norm of the components of the obtained RMSE_ϵ is taken, resulting in $\|\text{RMSE}_\epsilon\|$. The temporal division of the simulation is performed to highlight the performance of the algorithms in the presence of a dynamic maneuver, characterized by a maximum angular acceleration outside of $\pm 0.023 \text{ deg} \cdot \text{s}^{-2}$ in the initial segment of the simulation, and during a more stable attitude exhibiting a maximum angular acceleration within $\pm 0.023 \text{ deg} \cdot \text{s}^{-2}$, which resembles the tracking phase in the subsequent segment. The computed errors are collected in Table 2, where $\|\text{RMSE}_\epsilon\|$ represents the error derived from the estimation process itself for each algorithm, while RMSE_α gives the physical pointing error angle. As expected, the RMSE of the pointing error angle results in being lower than the norm of the RMSE of the attitude error vector for each scenario, the instantaneous angular misalignment being upper bounded by the attitude error magnitude [22].

The results of the entire simulation indicate that the quaternion propagation algorithm exhibits superior performance in comparison to the linear extrapolation. This observation is supported by the estimation errors for both $\|\text{RMSE}_\epsilon\|$ and RMSE_α with respect to the actual attitude of the ADCS satellite.

In the initial phases of the simulation, referred to as the dynamic phase, when the satellite angular velocity and angular acceleration are at their peak and the platform is undergoing significant movements, it becomes evident that the proposed quaternion propagation method, leveraging the LCT information, yields reduced errors in comparison to the linear extrapolation. This superior performance can be attributed to the fact that, in scenarios where attitudes undergo rapid and substantial changes, a linear function for extrapolation may be inadequate due to the nonlinear nature of the quantities being estimated. Moreover, the rapid fluctuations in the angular rate result in substantial variations in the attitude between successive updates.

Instead, the second part of the simulation can be considered as the tracking phase, during which the satellite maintains a stable attitude with respect to the Earth’s center while the attitude parameters undergo gradual changes in their absolute values. In this scenario, the linear extrapolation performs better. However, the performance of the quaternion propagation is hindered by the presence of noise, which affects the estimation through integration from the MEMS gyroscope sensor measurements. This noise imposes limitations on the minimum attainable estimation error, resulting in elevated errors for both the attitude error and the pointing error when compared to the errors obtained by the alternative method that does not utilize gyroscope sensor measurements.

Overall, it has already been demonstrated that the presence of any estimation process performed by the LCT at higher rates with respect to the satellite ADCS’s rate leads to a significant improvement in the attitude knowledge between satellite updates with respect to relying only on the information of the ADCS itself [25]. Nonetheless, the calculation of the next update based only on past measurements by linear extrapolation proved to be inaccurate when the attitude values change nonlinearly due to, e.g., a nonzero jerk in the acceleration profile. On the contrary, it was found that quaternion propagation algorithms can lead to a better attitude estimation until the lower limit of $0.4 \text{ deg} \cdot \text{s}^{-1}$ for the sinusoidal amplitude of Eq. (19), corresponding to an angular acceleration of more than or equal to $0.04 \text{ deg} \cdot \text{s}^{-2}$, after which this strategy should be employed with care.

Table 2 Performance parameters of the attitude estimation algorithms over the full simulation and for the dynamic and tracking scenarios

Parameter Algorithm	$\ \text{RMSE}_\epsilon\ $, deg			RMSE_α , deg		
	Full sim	Dynamic	Tracking	Full sim	Dynamic	Tracking
Linear extrapolation	0.3234	0.5114	0.0046	0.2349	0.3714	0.0035
Quaternion propagation	0.0262	0.0282	0.0248	0.0211	0.0226	0.0203

B. Hardware Validation Measurements

The testing setup, which comprises the necessary measurement and emulation equipment, contains the following devices. A telescope tube, referred to as the optical ground support equipment, simulates the beam emitted by an OGS as it would be received by the LCT in space. Further, it modulates the beacon signal with the same 10 kHz modulation frequency that is used to distinguish it from any background light. The LCT itself is mounted on a hexapod (HXP50-MECA, Newport), which can be precisely manipulated in all six degrees of freedom and controlled to mimic the behavior of the spacecraft. The tracking laser can be observed by an infrared camera, which serves as a reference to evaluate the performance of the algorithm.

To verify the performance of the estimation algorithm, the sensors must be calibrated first, as previously discussed in Sec. III.A. On the LCT, this concerns the MEMS gyroscope sensor and the MEMS FSM, which is coupled to the attitude of the satellite via the optical system. In both cases, the calibrated coordinate system of the ADCS, which is represented by the hexapod in the test, serves as a reference. In addition, the online bias compensation implementation is verified during a temperature cycle test.

1. Static Calibration for Device and Assembly Errors

In the test setup, the terminal is rotated on a turntable at multiple speeds around each axis. The used test runs include 15, −10, 5, −2.5, 1.25 and −10 deg·s^{−1} in that order, where each procedure is performed for approximately 10–20 s. The last movement with −10 deg·s^{−1} is used to finish the procedure at the starting point of the turntable. This ensures that the algorithm does not overfit the calibration parameters to one particular motion that is performed during the calibration process on orbit. The following calibration gain error matrix and offset for static calibration were obtained from the measurements according to Eq. (2):

$$C = \begin{bmatrix} 1.0008 & -0.0264 & 0.0184 \\ 0.0256 & 1.0052 & -0.0111 \\ -0.0556 & -0.0481 & -1.0066 \end{bmatrix}, \quad b = \begin{bmatrix} 0.35 \\ -2.69 \\ 0.28 \end{bmatrix} \quad (23)$$

Mounting the LCT into the satellite is limited to mechanical tolerances that impose deviations compared to the computer-aided design (CAD) coordinate frame. These variations can be accounted for by recording tracking data that contain mirror movements as well as gyroscope sensor measurements using an LSM similar to the initial static calibration but assuming that the sensor is already precalibrated. The remaining error should therefore only consist of coordinate rotations. Since the FSM has only two degrees of freedom and the gyroscope sensor measures in all three dimensions, the resulting rotation matrix is of size $R_{\text{est}} \in \mathbb{R}^{3 \times 2}$:

$$\hat{\omega}_{\text{gyro}}^{3 \times 1} = R_{\text{est}}^{3 \times 2} \cdot \omega_{\text{FSM}}^{2 \times 1} \quad (24)$$

The CAD model contains the theoretical rotation matrix between the gyroscope sensor, calibrated to the FSM frame, and the satellite's ADCS reference frame. Therefore, any deviation of the mounting orientation will be visible by comparing the rotation matrix R_{est} obtained from FSM-calibrated gyroscope sensor measurements with the theoretical rotation matrix R_{CAD} . The following results indicate a deviation from the ideal orientation:

$$R_{\text{est}} = \begin{bmatrix} 0.046 & 0.992 \\ 0.019 & -0.059 \\ 0.960 & -0.044 \end{bmatrix}, \quad R_{\text{CAD}} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \quad (25)$$

To validate the reconstruction of mounting errors, simulated data are purposely rotated and then compared with the original simulated data to determine the exact rotation parameters. Note that in order to model the real scenario of three gyroscope axes and two FSM axes, a matrix $R_{\text{est}} \in \mathbb{R}^{3 \times 2}$ has been estimated from the simulated data. By

calculating the cross-product between the two column vectors making up R_{est} , a third, perpendicular column vector emerges:

$$R_{\text{est},\perp} = R_{\text{est},1} \times R_{\text{est},2}, \quad R_{\text{est},3D} = [R_{\text{est},1}, R_{\text{est},2}, R_{\text{est},\perp}] \quad (26)$$

This vector is appended to the matrix to form $R_{\text{est},3D} \in \mathbb{R}^{3 \times 3}$, from which the angles can be determined. The sensor model used for the simulation is adapted from previous work [26] with added quantization, saturation, and erroneous influences such as scaling error, bias, and noise. Table 3 shows the angles set in the simulation and the reconstructed angular deviation, with α , β , and γ corresponding to a rotation around the x , y , and z axes in this order of the gyroscope sensor according to Fig. 5, respectively.

It can be seen that an accuracy of 15 mdeg or lower can be achieved.

2. Online Gyroscope Sensor Calibration During Dynamic Temperature Variations

The satellite will frequently experience temperature changes as it moves inside and outside of eclipse zones, which may lead to bias drifts. Keeping track of these drifts is essential to maintain good propagation quality. Instead of using a precalibrated temperature curve, the presented online bias estimation algorithm should be used to account for any deviation. To test the robustness of the online gyroscope sensor calibration algorithm to temperature changes, a hot air gun was used to heat the gyroscope sensor to approximately 50°C.

Figure 6 presents the results of an experiment where the hexapod remained stationary throughout, isolating temperature-induced effects. In Fig. 6a, noticeable shifts in estimated biases occur during the active heating and the following cooling phase, where the gyroscope sensor was allowed time to return to ambient temperature. Figure 6b demonstrates that the algorithm works effectively as the calibrated angular rates are close to a zero rate as expected. Note that the spikes are single-sample outliers from the measurement.

C. Hardware Attitude Knowledge Performance

The highly accurate encoders of the hexapod, integrated into the actuators, enabled the retrieval of a reference attitude, which was then used to assess the precision of the attitude propagation algorithm depicted in Fig. 3. The hexapod is commanded through a Python interface containing the relevant software drivers. This interface provides real-time readings of its attitude, which are fed to the CubeISL microcontroller via a serial interface to simulate the propagation reset with the estimated quaternions of the satellite ADCS.

1. Gyroscope-Sensor-Based Attitude Propagation

Table 4 illustrates the absolute error values in Euler angles for various propagation strategies, with resets occurring at 1 Hz. It presents the statistical results from the specific test and highlights the benefits of recursively calibrating the gyroscope sensor. The linear extrapolation condition refers to the scenario where the attitude is linearly extrapolated between ADCS estimates. The quaternion propagation refers to the case with the LCT gyroscope sensor involved. The online-limited method also accounts for scenarios where high accelerations would distort the online calibration, and related measurements are therefore automatically discarded.

Table 3 Comparison of real and reconstructed rotation angles with simulated data^a

Parameter	$\alpha_{x,\text{gyro}}$, deg	$\beta_{y,\text{gyro}}$, deg	$\gamma_{z,\text{gyro}}$, deg
Real angle	30	75	130
Reconstructed	29.992	75.007	130.015
Difference	0.008	0.007	−0.015

^aReconstruction is obtained from two axes of simulated data and a calculated perpendicular vector.

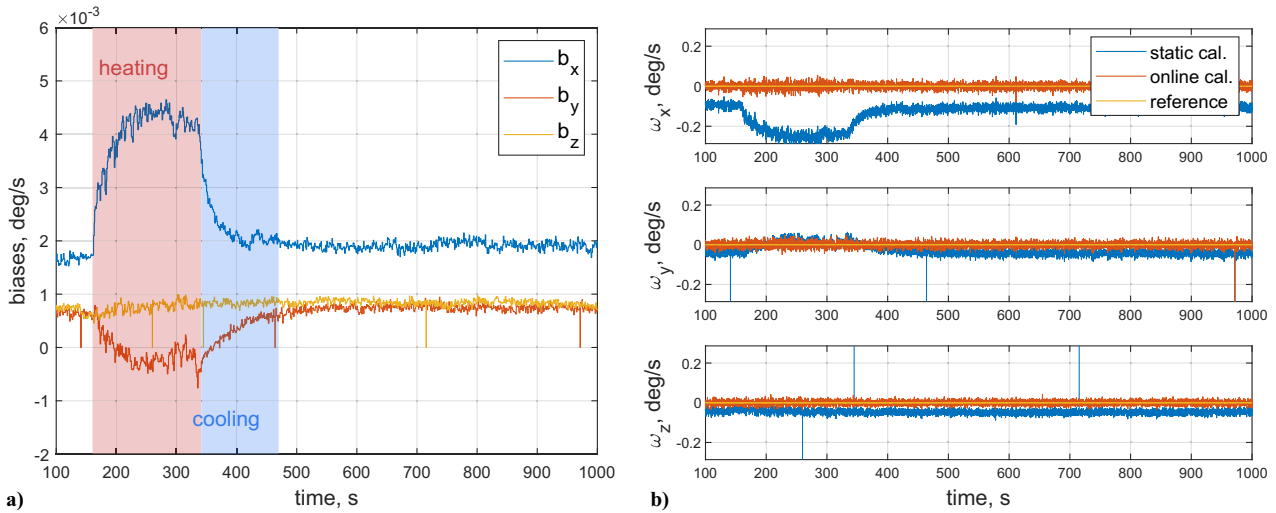


Fig. 6 a) Bias online estimation during temperature changes and b) comparison between an online and a statically calibrated gyroscope sensor during temperature changes.

Table 4 Error statistics definition of the linear extrapolation and quaternion propagation methods are as follows: linear propagation uses no measurements, while quaternion propagation uses internal gyroscope measurements^a

Parameter	μ_θ , deg	σ_θ , deg
Case 1: Linear extrapolation	0.0079	0.0141
Case 1: Quaternion propagation (static)	0.0200	0.0269
Case 1: Quaternion propagation (online)	0.0091	0.0153
Case 2: Linear extrapolation	0.7322	0.8533
Case 2: Quaternion propagation (online)	0.7003	0.8935
Case 2: Quaternion propagation (online-limited)	0.3625	0.4640

^a“Static” means only static calibration is applied, while “online” calibration is used with or without velocity limitations of the calibration reference.

Similar to the simulations, case 1 represents a scenario with slowly changing velocities comparable to the aforementioned tracking phase. Here, the linear extrapolation achieves comparable results with respect to the online-calibrated quaternion propagation. However, in the dynamic scenario of case 2, the online-limited quaternion propagation clearly outperforms the linear extrapolation. In duration with low dynamics, the online bias estimation has its

largest effect and can therefore be specifically used during those maneuvers, adding to the performance within accelerated rotations.

2. Feed-Forward Pointing Stability Compensation

Finally, the gained attitude knowledge of the LCT shall be used to compensate for the known pointing instabilities and drifts of the host platform. Therefore, hardware-in-the-loop tests using CubeISL are conducted to demonstrate that the FSM control with attitude information can augment the attitude knowledge and therefore reduce the acquisition time (see Fig. 7b). In Fig. 7a, an example of propagated offset angles due to body-pointing drift is depicted, showing how reference updates from the ADCS every second only have minor effects on the propagation and, therefore, the commanding of the FSM.

The hexapod’s initial position was set to simulate the scenario in which the received laser spot is at the center of the QPD, which resembles zero-error body pointing. Dynamic maneuvers within 1 deg and with $0.1 \text{ deg}\cdot\text{s}^{-1}$ angular velocity and $0.4 \text{ deg}\cdot\text{s}^{-2}$ angular acceleration are executed. These values represent the maximum field of regard offered by the FSM and the highest expected drift rate, respectively. In this way, it is directly visible whether the LCT is able to stay within the FOV of its tracking sensor.

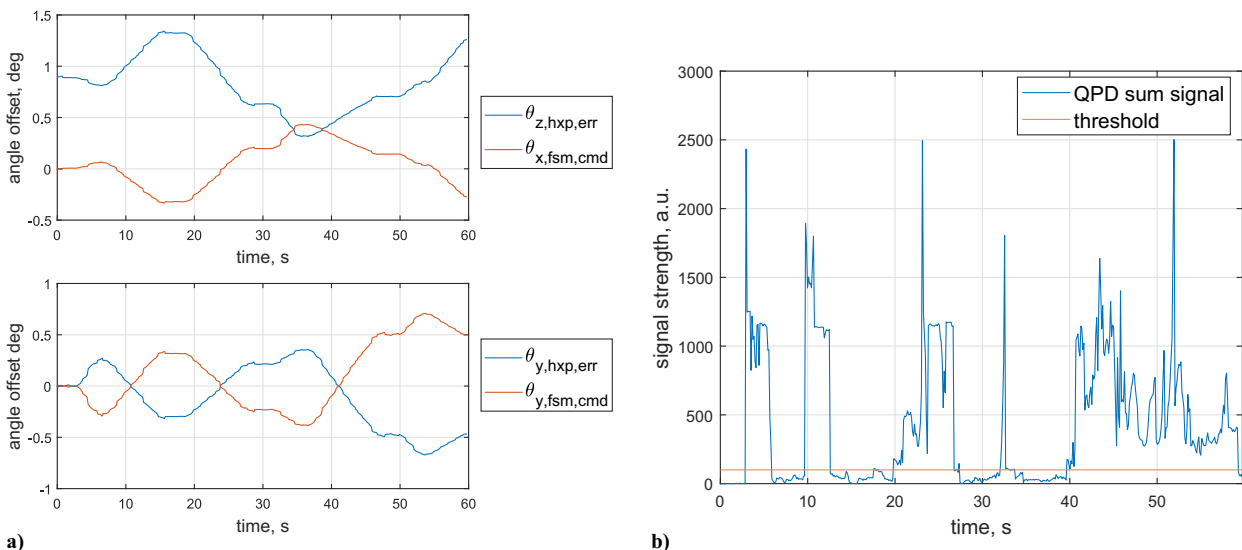


Fig. 7 a) Data of a quaternion propagated offset angle with feed-forward compensation commands to the FSM and b) sum of the QPD power sum signal with respect to the acquisition threshold as 16-bit integer values.

The results substantiate that, during the experiment, the laser remained within the FOV of the QPD for 63.3% of the time, meaning that the sum of the signal voltage exceeds the identified tracking threshold. This result shows that the compensation algorithm does not always compensate for the drift within the FOV, which can be explained by slight errors within the calibration and feed-forward path as well as measurement inaccuracies. However, as explained in the beginning, a search pattern will be superposed on the compensation. This means that remaining errors can be overcome. As a result of the active drift compensation, the pattern can be executed in a much smaller area, therefore increasing the success probability of link acquisition and reducing the expected time needed. This outcome confirms that using feed-forward compensation, the laser terminal can compensate for body pointing errors, allowing for increased acquisition performance due to a reduced FOU.

V. Conclusions

This study presents a feed-forward compensation approach that enhances the pointing accuracy of LCTs by integrating ADCS data with onboard gyroscope sensor measurements. By mitigating body-pointing uncertainties, the proposed method significantly improves the optical link acquisition process by mitigating satellite pointing errors using feed-forward compensation of the FSM, increasing the probability of successful link establishment while reducing acquisition time. In summary, the proposed methods are defined as follows: the linear extrapolation method does not use any measurements, whereas the quaternion propagation method uses measurements from internal gyroscopes. “Static” refers to calibration applied only once and remaining fixed, while “online” refers to continuous calibration, which may or may not include velocity limits for the calibration reference.

Hardware-in-the-loop validation and simulations demonstrate that quaternion propagation using gyroscopic data outperforms linear extrapolation, particularly in dynamic scenarios as in case 2. In case 1 (the low-dynamic tracking phase), where angular velocity changes gradually, both methods perform similarly. Quaternion propagation, which uses online calibration, achieves a mean error of 0.0091 deg and a standard deviation of 0.0153 deg. This is slightly higher than the mean error of 0.0079 deg and the standard deviation of 0.0141 deg achieved by linear extrapolation. However, in case 2 (high-dynamic scenario), where the satellite experiences rapid angular acceleration, quaternion propagation with an online-limited calibration achieves a mean error of 0.3625 deg and a standard deviation of 0.4640 deg, significantly outperforming linear extrapolation, which exhibits a mean error of 0.7322 deg and a standard deviation of 0.8533 deg. This represents a 50% improvement in mean error and a 45% reduction in pointing uncertainty under dynamic conditions.

The proposed method also provides an effective FOU reduction within ± 0.2 deg, ensuring that the FSM maintains optical alignment during acquisition even in the presence of body-pointing disturbances. Additionally, the online bias calibration technique successfully compensates for gyroscope sensor drift due to temperature variations, maintaining stable performance in varying environmental conditions. As an added benefit, pre- and postlaunch calibration can be used to detect any launch-related misalignment between the FSM and ADCS frames.

A key advantage of this approach is its ability to enhance pointing precision without requiring additional optical feedback during acquisition, making it highly suitable for CubeSats with size, weight, and power constraints. However, at angular accelerations below $0.04 \text{ deg}\cdot\text{s}^{-2}$, gyroscope sensor noise and calibration errors start to degrade performance, limiting its applicability in low-dynamic scenarios such that linear extrapolation is preferred.

Appendix: Quaternion Multiplication

As noted in [22], quaternion multiplication can be expressed using matrix operations, analogous to the vector cross-product:

$$\mathbf{q} \otimes \bar{\mathbf{q}} = [\mathbf{q} \otimes] \bar{\mathbf{q}} = \bar{\mathbf{q}} \odot \mathbf{q} \quad (\text{A1a})$$

$$\mathbf{q} \odot \bar{\mathbf{q}} = [\mathbf{q} \odot] \bar{\mathbf{q}} = \bar{\mathbf{q}} \otimes \mathbf{q} \quad (\text{A1b})$$

where

$$[\mathbf{q} \otimes] \equiv \begin{bmatrix} q_4 I_{3 \times 3} - [\mathbf{q}_{1:3}]_{\times} & \mathbf{q}_{1:3} \\ -\mathbf{q}_{1:3}^T & q_4 \end{bmatrix} = [\Psi(\mathbf{q}) \quad \mathbf{q}] \quad (\text{A2})$$

and

$$[\mathbf{q} \odot] \equiv \begin{bmatrix} q_4 I_{3 \times 3} + [\mathbf{q}_{1:3}]_{\times} & \mathbf{q}_{1:3} \\ -\mathbf{q}_{1:3}^T & q_4 \end{bmatrix} = [\Xi(\mathbf{q}) \quad \mathbf{q}] \quad (\text{A3})$$

with $\Psi(\mathbf{q})$ and $\Xi(\mathbf{q})$ being the 4×3 matrices:

$$\Psi(\mathbf{q}) \equiv \begin{bmatrix} q_4 I_{3 \times 3} - [\mathbf{q}_{1:3}]_{\times} \\ -\mathbf{q}_{1:3}^T \end{bmatrix} = \begin{bmatrix} q_4 & q_3 & -q_2 \\ -q_3 & q_4 & q_1 \\ q_2 & -q_1 & q_4 \\ -q_1 & -q_2 & -q_3 \end{bmatrix} \quad (\text{A4})$$

$$\Xi(\mathbf{q}) \equiv \begin{bmatrix} q_4 I_{3 \times 3} + [\mathbf{q}_{1:3}]_{\times} \\ -\mathbf{q}_{1:3}^T \end{bmatrix} = \begin{bmatrix} q_4 & -q_3 & q_2 \\ q_3 & q_4 & -q_1 \\ -q_2 & q_1 & q_4 \\ -q_1 & -q_2 & -q_3 \end{bmatrix} \quad (\text{A5})$$

Acknowledgments

This research work is supported by the German Federal Ministry of Defense through the technological research and development assignment “Responsive Space Capabilities.” The TDK InvenSense sensors described in this publication were not designed, tested, or qualified for aerospace use. TDK InvenSense makes no representations or warranties regarding their suitability, reliability, or performance in such environments. Any use of these products in aerospace applications is entirely at the user’s own risk. TDK InvenSense does not object to the research use of these products as described in this publication, but this does not constitute an endorsement or recommendation of these products in aerospace or space applications. During the preparation of this work, the authors used DeepL in order to improve the readability and language of the paper.

References

- [1] Weston, S. V., Burkhard, C. D., Stupl, J. M., Ticknor, R. L., Yost, B. D., Austin, R. A., Galchenko, P., Newman, L. K., and Santos Soto, L., “State-of-the-Art Small Spacecraft Technology,” NASA TP 20250000142, Washington, D.C., 2025, <https://www.nasa.gov/smallsat-institute/sst-soa/>.
- [2] Elsner, L., Petermann, T., von Arnim, M., and Schilling, K., “Pre-Flight Verification of the Cubesat Attitude Control System for the QUBE Mission,” *4S Symposium 2024*, 2024, Paper 155. <https://doi.org/10.1117/12.3062785>
- [3] Kingsbury, R., Riesing, K., and Cahoy, K., “Design of a Free-Space Optical Communication Module for Small Satellites,” *Proceedings of the Small Satellite Conference*, 2014, SSC14-IX-6, <https://digitalcommons.usu.edu/smallsat/2014/AdvTechComm/6/>.
- [4] Agrawal, P. S., Fuller, C. O., Kung, K. R., and DiMatteo, J. J., “From Design to Orbit: Engineering Sub-Arsecond CubeSat Pointing Performance on Pathfinder-3,” *Proceedings of the Small Satellite Conference*, 2024, SSC24-IX-02, <https://digitalcommons.usu.edu/smallsat/2024/all2024/125/>.
- [5] Li, S., and Zhong, M., “High-Precision Disturbance Compensation for a Three-Axis Gyro-Stabilized Camera Mount,” *IEEE/ASME Transactions on Mechatronics*, Vol. 20, No. 6, 2015, pp. 3135–3147. <https://doi.org/10.1109/TMECH.2015.2394456>

- [6] Dell'Olio, F., Natale, T., Wang, Y.-C., and Hung, Y.-J., "Miniaturization of Interferometric Optical Gyroscopes: A Review," *IEEE Sensors Journal*, Vol. 23, No. 24, 2023, pp. 29948–29968. <https://doi.org/10.1109/JSEN.2023.3327217>
- [7] Heimann, M., Liesegang, M., Arndt-Staufenbiel, N., Schröder, H., and Lang, K.-D., "Optical System Components for Navigation Grade Fiber Optic Gyroscopes," *Emerging Technologies in Security and Defence; and Quantum Security II; and Unmanned Sensor Systems X, SPIE*, 2013, Paper 88991A. <https://doi.org/10.1117/12.2029097>
- [8] Shukri, Z., Rezaei, H., Fasihanifard, M., Ghofrani, S., Bissonnette, J., Gravey, T., Romanuik, S., Tewari, R., and Zandi, K., "Photonic Integrated Circuit Microchip-Based Optical Gyroscopes for High-Precision Inertial Measurement Units," *Integrated Optics: Devices, Materials, and Technologies XXIX*, edited by S. M. García-Blanco, and P. Cheben, SPIE, 2025, p. 25. <https://doi.org/10.1117/12.3041353>
- [9] Rüddenklau, R., and Schitter, G., "Optimisation of Acquisition Patterns for Establishing Inter CubeSat Optical Communications," *Journal of Optical Communications and Networking*, Vol. 16, No. 8, 2024, Paper 814. <https://doi.org/10.1364/JOCN.518004>
- [10] AAC Clyde Space, "iADCS400," 2024, <https://www.aac-clyde.space/wp-content/uploads/2021/11/iADCS400-1.pdf>
- [11] Alén Space, "6U Platform," 2024, <https://alen.space/products/6u-platform/>
- [12] Blue Canyon, "XB6," 2024, <https://www.bluecanyontech.com/wp-content/uploads/6U-2026.pdf>
- [13] Gomspace, "6U Premium," 2024, <https://gomspace.com/6u-premium.aspx>
- [14] Nano Avionics, "M6P Max," 2024, <https://nanoavionics.com/small-satellite-buses/6u-cubesat-nanosatellite-m6p/>
- [15] Rüddenklau, R., Rein, F., Roubal, C., Rödiger, B., and Schmidt, C., "In-Orbit Demonstration of Acquisition and Tracking on OSIRIS4CubeSat," *Optics Express*, Vol. 32, No. 23, 2024, pp. 41,188–41,200. <https://doi.org/10.1364/OE.537889>
- [16] Nonay, J. R., Rüddenklau, R., Sinn, A., Jakobs, J. P., Berlitz, J., Rödiger, B., and Schitter, G., "Horizontal Free-Space Optical Link with CubeISL over 143 km," *Journal of Optical Communications and Networking*, Vol. 16, No. 5, 2024, pp. 593–601. <https://doi.org/10.1364/JOCN.518271>
- [17] Bangert, P., Busch, S., and Schilling, K., "In-Orbit Performance of the Pico-Satellite UWE-3," 2015, https://www.researchgate.net/profile/Stephan-Busch-4/publication/286379222_In-Orbit_Performance_of_the_Pico-Satellite_UWE-3/links/614f4faf522ef665fb5ec793/In-Orbit-Performance-of-the-Pico-Satellite-UWE-3.pdf
- [18] Poddar, S., Kumar, V., and Kumar, A., "A Comprehensive Overview of Inertial Sensor Calibration Techniques," *Journal of Dynamic Systems, Measurement, and Control*, Vol. 139, No. 1, 2017, Paper 011006. <https://doi.org/10.1115/1.4034419>
- [19] Springmann, J. C., and Cutler, J. W., "Attitude-Independent Magnetometer Calibration with Time-Varying Bias," *Journal of Guidance, Control, and Dynamics*, Vol. 35, No. 4, 2012, pp. 1080–1088. <https://doi.org/10.2514/1.56726>
- [20] Springmann, J. C., and Cutler, J. W., "Flight Results of a Low-Cost Attitude Determination System," *Acta Astronautica*, Vol. 99, June–July 2014, pp. 201–214. <https://doi.org/10.1016/j.actaastro.2014.02.026>
- [21] Wertz, J. R., *Spacecraft Attitude Determination and Control*, Vol. 73, Springer Science & Business Media, Berlin, 2012, pp. 446–459.
- [22] Markley, F. L., and Crassidis, J. L., *Correction to: Fundamentals of Spacecraft Attitude Determination and Control*, Springer, Berlin, 2014. <https://doi.org/10.1007/978-1-4939-0802-8>
- [23] Milanovic, V., Matus, G. A., and McCormick, D. T., "Gimbal-Less Monolithic Silicon Actuators for Tip–Tilt–Piston Micromirror Applications," *IEEE Journal of Selected Topics in Quantum Electronics*, Vol. 10, No. 3, 2004, pp. 462–471. <https://doi.org/10.1109/JSTQE.2004.829205>
- [24] TDK InvenSense, "IAM20380," 2024, https://invensense.tdk.com/wp-content/uploads/2022/09/DS-000195-IAM-20380_v1.1-Typ.pdf
- [25] Garbagnati, E., Rüddenklau, R., and Schmidt, C., "Augmenting Attitude Knowledge for Optical Communication Terminals on CubeSats by Kalman Filter Based Estimation," *Proceedings of the Small Satellite Conference*, 2025, SSC25-RAVIII-06, <https://digitalcommons.usu.edu/smallsat/2025/RA-S8-2025/4/>
- [26] Kwon, H. J., Seok, S., and Lim, G., "System Modeling of a MEMS Vibratory Gyroscope and Integration to Circuit Simulation," *Sensors*, Vol. 17, No. 11, 2017, Paper 2663. <https://doi.org/10.3390/s17112663>

Y. Cheng
Associate Editor